

AI Critique — Candidate Draft

Status: **PENDING HUMAN REVIEW — NOT YET THE STUDENT'S FINAL CRITIQUE**

Word count: **278 words** (whitespace count of the substantive body)

AI accelerated this assignment by tracing the source code, mapping the WF-03 API lifecycle, generating a shared k6 architecture, and parsing thousands of real measurements reproducibly. However, several failures showed why its output could not be accepted without human review. The first draft used `::` inside k6 group/check names, which k6 v2.2.0 rejects. Its generic exception handler then allowed a tight retry loop, creating about 9.7 million iterations and roughly 21.6 GiB of non-HTTP output. It also initially mislabeled Pilot custom metrics as measured traffic and invoked the provisioning helper from the wrong trust boundary. Later, a non-fail-closed listener check allowed setup accounts to reach an unrelated localhost SUT process. These were test-harness defects, not SUT performance findings.

AI interpretation also required correction. Checkout and cancellation had the highest relative latency, but their millisecond-scale values and zero failures did not prove bottlenecks. Proposed p95, lifecycle, success-rate, and regression margins lacked repeat-run noise evidence, so I classified them as insufficiently supported rather than final thresholds. Generic recommendations for a normal index on `LIKE '%term%'` search and a connection pool for a local SQLite object were hallucinated or inapplicable.

Human review improved the work by introducing dedicated per-VU accounts, realistic cancellation think time, iteration-level authentication failure, exact process ownership guards, bounded output, and immutable failure evidence. The main lesson is that AI is strongest as a fast investigator and automation partner, while the student must own semantics, safety, and conclusions. Small Pilots, raw-data verification, and explicit human checkpoints converted plausible drafts into defensible performance evidence without hiding mistakes. The collaboration also showed that auditability matters: preserving failed runs made later safety improvements explainable, reproducible, and distinguishable from genuine SUT behavior.

The student must edit or explicitly approve this substantive text before it is used as the final mandatory critique.